EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Midterm Examination, Summer 2025

Course: CSE 106 Discrete Mathematics, Section-03 **Date:** July 30, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor, CSE Department
Full Marks: 60 (Will be converted to 30 and applied towards final grading)
Time: 1 hour 20 Minutes

Note: There are **TEN** questions, answer **ALL** of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

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1. Let p, q, and r be the propositions:

- p: Grizzly bears have been seen in the area.
- q: Hiking is safe on the trail.
- r: Berries are ripe along the trail.

Write these propositions using p, q, r, and logical connectives (including negations).

(a) Berries are ripe along the trail, but grizzly bears have not been seen in the area.

(b) If berries are ripe along the trail, hiking is safe if and only if grizzly bears have not been seen in the area.

(c) Hiking is not safe on the trail whenever grizzly bears have been seen in the area and berries are ripe along the trail.

[CO1, C2] [Mark: 6]

2. Determine whether each of the following logical equivalences is **true** or **false**.

(a) $\neg(p \wedge q) \equiv \neg p \vee \neg q$

(b) $p \rightarrow q \equiv \neg q \rightarrow p$

(c) $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$

(d) $\neg(p \rightarrow q) \equiv p \vee \neg q$

[CO1, C3] [Mark: 8]

3. Determine the truth value of each of the following statements if the domain for all variables consists of all integers.

(a) $\exists n\,(n^2 = 2)$

(c) $\exists n\,(n^2 < 0)$

(b) $\forall n\,(n^2 \geq n)$

(d) $\exists n\,(n < 0 \wedge n^2 > 100)$

[CO1, C3] [Mark: 8]

4. Construct the truth table for the compound proposition.

$$(p \vee q) \rightarrow (p \wedge \neg q)$$

[CO1, C2, Mark: 8]

5. Show that if n is an integer and $n^3 + 5$ is odd, then n is even.

[CO1, C3, Mark: 4]

6. Use mathematical induction to prove the following statement for all integers $n \geq 0$:

$$1 + 3 + 3^2 + 3^3 + \cdots + 3^n = \frac{3^{n+1} - 1}{2}$$

[CO1, C3, Mark: 6]

7. Let $A = \{5, 8, 9\}$ and $B = \{x, y, z\}$.

(a) List all elements of the Cartesian product $A \times B$.

(b) Determine the cardinality of the power set $\mathcal{P}(A \times B)$.

[CO4, C2, Mark: 4]

8. Determine whether the following functions is a bijection from R to R.

$$f(x) = x^4 + 5$$

[CO4, C2, Mark: 6]

9. Find $f(2)$, $f(3)$, $f(4)$, and $f(5)$ if f is defined recursively by

$$f(0) = -1, f(1) = 2$$
$$f(n+1) = f(n) + 2f(n-1)$$

[CO4, C3, Mark: 4]

10. Assume $R : A \rightarrow A$ is a relation where $A = \{1, 2, 3, 4\}$. R is given by $\{(x, y) : x \in A, y \in A, xy < 10\}$. Determine whether the following statements are true or false with justification.

(a) R is reflexive.

(b) R is symmetric.

(c) R is transitive.

[CO4, C3, Mark: 6]

EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Midterm Examination, Summer 2025

Course: CSE 106 Discrete Mathematics, Section-09 **Date:** August 2, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor, CSE Department
Full Marks: 60 (Will be converted to 30 and applied towards final grading)
Time: 1 hour 20 Minutes

Note: There are **TEN** questions, answer **ALL** of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

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1. Let p, q, and r be defined as follows. Write the propositions in (a), (b), and (c) using p, q, r, and logical connectives (including negations). Only provide answers.

- p: You get an A on the final exam.
- q: You do every exercise in this book.
- r: You get an A in this class.

(a) You get an A in this class, but you do not do every exercise in this book.

(b) You get an A on the final, you do every exercise in this book, and you get an A in this class.

(c) To get an A in this class, it is necessary for you to get an A on the final.

[CO1, C2] [Mark: 6]

2. Use a truth table to verify the following **De Morgan's Law** in propositional logic.

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

[CO1, C2] [Mark: 8]

3. . Determine the truth value of each of these statements with appropriate justification if the domain consists of all real numbers.

(a) $\exists n\,(n^3 = -1)$ (b) $\exists n\,(n^5 \leq n^2)$ (c) $\forall n\,(n^2 > n)$ (d) $\exists n\,(n^2 + 2) < 3$

[CO1, C3] [Mark: 8]

4. State the converse and contrapositive of each of the following conditional statements.

(a) If it snows tonight, then I will stay at home.

(b) I go to the beach whenever it is a sunny summer day.

(c) When I stay up late, it is necessary that I sleep until noon.

(d) I come to class whenever there is going to be a quiz.

[CO1, C3, Mark: 8]

5. Show that if n is an integer and $5n + 6$ is odd, then n is odd.

[CO1, C3, Mark: 4]

6. Use mathematical induction to prove the following statement for all integers $n \geq 1$:

$$1.1! + 2.2! + \cdots + n.n! = (n + 1)! - 1$$

[CO1, C3, Mark: 6]

7. List the members of the following sets.

(a) $\{x : x \text{ is a real number such that } x^2 = 1\}$

(b) $\{x : x \text{ is a positive integer less than } 12\}$

(c) $\{x : x \text{ is the square of an integer and } x < 100\}$

(d) $\{x : x \text{ is an integer such that } x^2 = 2\}$

[CO4, C2, Mark: 4]

8. Let $f(x) = x^2 + 1$ and $g(x) = x + 2$. Find the value of the following:

(a) $(f \circ g)(4)$ (b) $(g \circ f)(7)$ (c) $(f \circ g)(5)$

[CO4, C2, Mark: 6]

9. Find $f(2)$, $f(3)$, $f(5)$, and $f(6)$ if f is defined recursively by

$$f(0) = 2, f(1) = 3$$
$$f(n + 1) = f(n)f(n - 1)$$

[CO4, C3, Mark: 4]

10. Assume $R : A \to A$ is a relation where $A = \{1, 3, 5, 7\}$. R is given by $\{(x, y) : x \in A, y \in A, |x - y| < 4\}$. Determine whether the following statements are true or false with justification.

(a) R is irreflexive.

(b) R is anti-symmetric.

(c) R is transitive.

[CO4, C3, Mark: 6]

EAST WEST UNIVERSITY

Department of Computer Science and Engineering

B.Sc. in Computer Science and Engineering Program

Midterm Examination, Summer 2025

Course: CSE 106 Discrete Mathematics, Section-10 **Date:** August 2, 2025
Instructor: Dr. Nafis Sadeq, Assistant Professor, CSE Department
Full Marks: 60 (Will be converted to 30 and applied towards final grading)
Time: 1 hour 20 Minutes

Note: There are **TEN** questions, answer **ALL** of them. Course Outcome (CO), Cognitive Level, and Mark of each question are mentioned at the right margin.

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1. How many rows appear in a truth table for each of these compound propositions?

(a) $p \wedge q \wedge r$

(b) $(p \vee q) \rightarrow (r \vee s)$

(c) $(p \leftrightarrow q) \vee (\neg r \wedge s) \vee (t \wedge \neg u)$

[CO1, C2] [Mark: 6]

2. Use a truth table to show that $\neg(p \rightarrow q)$ and $p \wedge \neg q$ are logically equivalent.

[CO1, C2] [Mark: 8]

3. Let $Q(x)$ be the statement $x + 1 > 2x$. If the domain consists of all integers, what are these truth values?

(a) $Q(1)$ (b) $\exists x Q(x)$ (c) $\exists x \neg Q(x)$ (d) $\forall x \neg Q(x)$

[CO1, C3] [Mark: 8]

4. Let p and q be the propositions:

- p: I bought a lottery ticket this week.
- q: I won the million dollar jackpot.

Express each of these propositions as an English sentence.

(a) $\neg p$ (b) $p \vee q$ (c) $p \iff q$ (d) $\neg p \rightarrow \neg q$

[CO1, C3, Mark: 8]

5. Prove that if n is a positive integer, then n is even if and only if $11n + 8$ is even.

[CO1, C3, Mark: 4]

6. Prove that for every positive integer n.

$$1.2 + 2.3 + \cdots + n(n+1) = n(n+1)(n+2)/3$$

[CO1, C3, Mark: 6]

7. Three sets A, B, C are defined as follows.

$$A = \{x \in \mathbb{N} \mid 1 \leq x \leq 3\}$$
$$B = \{x \in \mathbb{N} \mid x = 3k,\ 1 \leq k \leq 4\}$$
$$C = \{x \in \mathbb{N} \mid x = 5k,\ 1 \leq k \leq 5\}$$

Using these definitions, compute the following:

(a) The cardinality of the power set $\mathcal{P}(A \times B)$

(b) The cardinality of the power set $\mathcal{P}(B \times C)$

[CO4, C2, Mark: 4]

8. Determine whether the following functions is a onto function from $\mathbb{R}$ to $\mathbb{R}$.

$$f(x) = 3x^2 - 4$$

[CO4, C2, Mark: 6]

9. Find $f(1)$, $f(2)$, $f(3)$, and $f(4)$ if f is defined recursively by

$$f(0) = 2$$
$$f(n+1) = 4f(n) + 3$$

[CO4, C3, Mark: 4]

10. Assume $R : A \to A$ is a relation where $A = \{1, 2, 3, 4\}$. R is given by $\{(x, y) : x \in A, y \in A, 2x + y < xy\}$. Determine whether the following statements are true or false with justification.

(a) R is reflexive.

(b) R is symmetric.

(c) R is transitive.

[CO4, C3, Mark: 6]